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Department of Nano Science Technology
Faculty of Technology

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Centralized Workshops & Lab Scheduling Platform

Database SQL Query Sheet

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Workshop Management System - Database Design & Implementation Documentation

The Workshop Management System is developed using C++ with the Qt Framework, utilizing SQLite as an embedded relational database backend.

1. Relational Schema and Data Dictionary

1.1 Students Table (Parent Relation)

Column Name	Data Type	Constraint / Key	Description
Index_No	TEXT	PRIMARY KEY	Unique student registration identifier (e.g., 'EG/2022/1001')
Name	TEXT	NOT NULL	Full name of the student
Dep	TEXT	NOT NULL	Academic department of study (e.g., Mechanical, Electrical, Materials)
Email	TEXT	NOT NULL, UNIQUE	Official student email address
PDF_Path	TEXT	DEFAULT NULL	Relative server file path to uploaded consent/safety forms

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1.2 Bookings Table (Child Relation)

Column Name	Data Type	Constraint / Key	Description
Booking_ID	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique auto-generated surrogate key for reservation tracking
Index_No	TEXT	FOREIGN KEY, NOT NULL	References Students(Index_No) with ON DELETE/UPDATE CASCADE
Date	TEXT	NOT NULL	Scheduled reservation date (ISO-8601: YYYY-MM-DD)
Time	TEXT	NOT NULL	Scheduled start time (24-hour format: HH:MM:SS)
Description	TEXT	NOT NULL	Purpose of access and equipment required
Admin_Text	TEXT	DEFAULT NULL	Supervisor/Technical officer remarks or rejection reasons
Status_Option	TEXT	DEFAULT 'Pending', CHECK constraint	State restricted to ('Pending', 'Approved', 'Rejected')
Created_At	DATETIME	DEFAULT CURRENT_TIMESTAMP	Audit timestamp of booking creation

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2. SQL Implementation Script (DDL & DML)

```
-- 1. Enable foreign key support for SQLite
PRAGMA foreign_keys = ON;

-- Drop existing tables to ensure clean deployment
DROP TABLE IF EXISTS Bookings;
DROP TABLE IF EXISTS Students;

-- 2. CREATE TABLE Statements (DDL)

-- Student details table (Parent)
CREATE TABLE Students (
    Index_No TEXT PRIMARY KEY,
    Name TEXT NOT NULL,
    Dep TEXT NOT NULL,
    Email TEXT NOT NULL UNIQUE,
    PDF_Path TEXT DEFAULT NULL
);

-- Workshop machine bookings table (Child)
CREATE TABLE Bookings (
    Booking_ID INTEGER PRIMARY KEY AUTOINCREMENT,
    Index_No TEXT NOT NULL,
    Date TEXT NOT NULL,
    Time TEXT NOT NULL,
    Description TEXT NOT NULL,
    Admin_Text TEXT DEFAULT NULL,
    Status_Option TEXT DEFAULT 'Pending'
        CHECK (Status_Option IN ('Pending', 'Approved', 'Rejected')),
    Created_At DATETIME DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_bookings_students
        FOREIGN KEY (Index_No) REFERENCES Students(Index_No)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

-- Index for foreign key lookup optimization
CREATE INDEX idx_bookings_student ON Bookings(Index_No);

-- 3. Sample Data Insertion (DML)

-- Sample data: Students
INSERT INTO Students (Index_No, Name, Dep, Email, PDF_Path) VALUES
('EG/2022/1001', 'Kamal Perera', 'Mechanical', 'kamal.p@univ.lk',
'uploads/EG20221001_form.pdf'),
('EG/2022/1045', 'Nimal Silva', 'Electrical', 'nimal.s@univ.lk',
'uploads/EG20221045_form.pdf'),
```

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```
('EG/2022/1120', 'Saman Kumara', 'Materials', 'saman.k@univ.lk', NULL);

-- Sample data: Bookings
INSERT INTO Bookings (Index_No, Date, Time, Description, Admin_Text,
Status_Option) VALUES
('EG/2022/1001', '2026-09-25', '09:30:00', 'Lathe Machine access for final year
project', 'Approved with safety gear', 'Approved'),
('EG/2022/1045', '2026-09-26', '14:00:00', 'PCB Milling Machine testing', NULL,
'Pending'),
('EG/2022/1001', '2026-09-28', '11:00:00', 'Welding unit access for chassis
fabrication', 'Supervisor signature missing', 'Rejected');
```